CUDA Graph for Head Pipelining

All-head replay cuts launch gaps; head bands expose overlap

Request-stable shape

- One video keeps the same H/W/T and reference layout through denoising

All heads + Graph

- Cuts launch overhead, but attention still waits for the full exchange

Head bands + Graph

- Starts attention as each fixed band arrives and pipelines later bands

Capture boundary

- Graph: unpack → attention → pack; RDMA and readiness events stay outside



Same Graph Benefit · Different Overlap

Head batching exposes overlap; CUDA Graph keeps the finer pipeline from becoming launch-bound

Sources: CUDA Programming Guide · current H3 Ulysses prototype execution contract · author diagram